

GEOMETRY	b52
OBJECTIVE	Solve the drag on this airframe and paint the surface pressure field.
ISSUED (UTC)	2026-07-21T18:04:00Z

RESULT

Drag coefficient 0.0471

TREND ONLY

Envelope: 0.0018
Trust basis: steady RANS on a watertight airframe with no in-regime experimental anchor for this configuration; the reference area is the measured silhouette (601 m^2), not the published wing area

Lift coefficient	0.0192	0.0041
Mesh	193,880 cells	max non-orthogonality 62, max skewness 3.1

UNCERTAINTY CHANNELS - ASME V&V 20

Input	48.5 m ref. length	(quantified)
	reference length stated, not inferred; freestream at cruise	
Numerical	grid difference	(not quantified)
	single mesh; a two-mesh probe yields a grid difference, not a verified GCI (Eca & Hoekstra)	
Model form	fully-turbulent RANS	(not quantified)
	k-omega SST, no transition model; Aref is the measured silhouette, so Cd is not directly comparable to published book values	

COMPUTE	193,880 cells - 8.0 core-min spent
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EVIDENCE-BUNDLE SEAL - SHA-256
025c221e1e43fcb26487a4ad89bce43
a419210658f8af3e9ff6598597778e9b

Sealed by the hash above; any change to the recorded run invalidates it. No tier exceeds TREND ONLY without an experimental comparison.